



Title:

RESTAURANT MANAGEMENT SYSTEM

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OVERVIEW

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Project Overview

The Restaurant Management System is a console-based application developed in C to manage the core activities of a small restaurant.

The system manages:

- **Customer order placement**
- **Kitchen order processing**
- **Menu administration**
- **Order tracking**
- **Sales and order reports**

OBJECTIVES

Functional Objectives:

- Develop a menu-driven restaurant application.
- Allow customers to view menus and place orders.
- Manage orders using appropriate data structures.
- Process orders in the sequence received.
- Allow administrators to manage menu items.
- Calculate customer bills automatically.
- Generate active, completed, and cancelled order reports .

Technical Objectives:

Apply C programming concepts through
Structures + Linked Lists + Queue
+ Stack + Sorting + Functions

SYSTEM ARCHITECTURE

Three Functional Modules:

- Customer
- Kitchen
- Admin

```
"C:\Users\Acer\Downloads\m" X + v
-----
RESTAURANT MANAGEMENT SYSTEM
-----
1. Customer
2. Kitchen
3. Admin
4. Exit
-----
Choose an option: |
```

Module-I: CUSTOMER

Main Functions;

- View available food items.
- Select items using Item ID.
- Enter required quantity.
- Automatically calculate total bill.
- Search active orders using Order ID.
- Cancel an order before preparation.

```
-----  
CUSTOMER MENU  
-----  
1. View Menu  
2. Place Order  
3. Search Order  
4. Cancel Order  
5. Back  
-----  
Choose an option: |
```

ID	ITEM	PRICE
1	Chicken Burger	5.50
2	Beef Steak	12.75
3	Cheese Pizza	8.00
4	Veg Salad	3.25
5	Cold Coffee	2.50
6	Vanilla Ice Cream	2.00

6 item(s)

Module-2: KITCHEN

Main Functions:

- View Waiting Orders
- FIFO Processing
- Prepare Next Order

KITCHEN MENU

1. View Waiting Orders
2. Prepare Next Order
3. Back

> Order #1001 prepared and served.

Order #1001 Customer: ariyana [Completed]

Chicken Burger	1 x	5.50 =	5.50
Beef Steak	2 x	12.75 =	25.50
TOTAL			31.00

Module-3: ADMIN

Main Functions;

- Management & Reporting
- Manage Menu
- Sort Menu
- View Reports

ADMIN MENU

1. Add Menu Item
2. Delete Menu Item
3. Update Menu Item
4. Display Menu
5. Sort Menu by Price (Bubble Sort)
6. Reports
7. Back

> Menu sorted by price.

ID	ITEM	PRICE
6	Vanilla Ice Cream	2.00
5	Cold Coffee	2.50
4	Veg Salad	3.25
1	Chicken Burger	5.50
3	Cheese Pizza	8.00
2	Beef Steak	12.75

DATA STRUCTURE

01 — Linked List

Active Orders

- Insert order
- Search order
- Remove order

03 — Stack

Order History — LIFO

- Completed orders
- Cancelled orders

02 — Queue

Kitchen Processing — FIFO

- Enqueue
- Dequeue
- Remove waiting order

ALGORITHMS

Algorithms Used	Purpose	HOW IT WORKS
Bubble Sort	Sort menu items by price.	Compare → Swap → Repeat
Sequential Search	Locate menu items and orders.	Check → Compare → Find
Recursion	Used to count: Completed orders,Cancelled orders	

VALIDATION & MEMORY

Input Validation

- Empty names rejected
- Invalid IDs rejected
- Quantity: 1–100
- Price must be greater than 0
- Numeric inputs validated

Memory Management

`malloc()` → Dynamic Allocation
`freeAll()` → Memory Cleanup

Strengths and Limitations

Strengths

- Automatic bill calculation
- FIFO-based kitchen processing
- Organized order history
- Menu administration
- Useful reports
- Multiple data structures in one system
- Modular design

Limitations

- No permanent data storage
- No login system
- Console-only interface
- Fixed maximum menu size
- Orders cannot be modified after placement
- Single-user operation

Conclusions

Overall, the project provides practical experience in integrating multiple data structures into a single working application and creates a foundation for future improvements such as database storage, GUI support, login systems, and advanced reporting.



THANK YOU